Task 3: Probability and Congestion Analysis

Define **congestion** as: traffic volume > 5,500 vehicles.

Calculate:

3.1 Basic probability

- **P(Congestion)**
- **P(Clear Weather)**
- **P(Congestion AND Clear Weather)**

3.2 Conditional probability

Calculate:

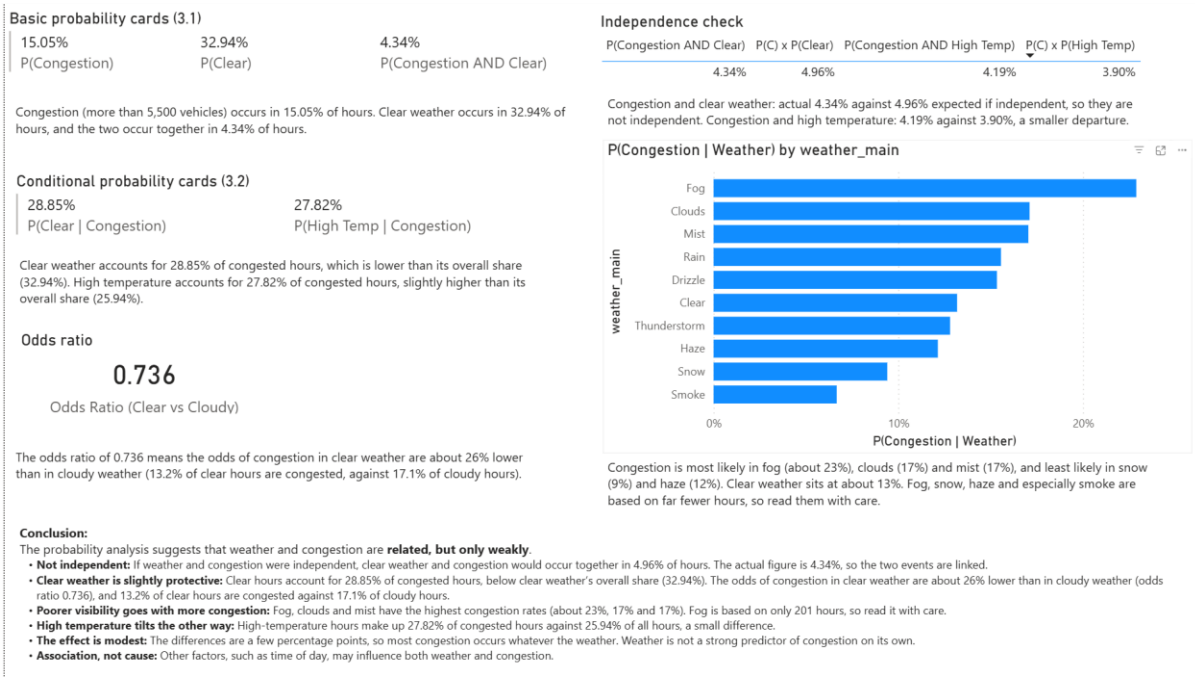
- **P(Clear Weather | Congestion)**
- **P(High Temperature | Congestion)**, where high temperature > 292K
- Determine whether weather conditions and congestion can be considered independent using $P(A \cap B) = P(A) \times P(B)$
- Calculate the **odds ratio of congestion in clear versus cloudy weather**.

Conclude with a short explanation of what the probability analysis suggests about the relationship between weather and congestion.

Solution:

I solved this using SQL queries but while the output was the same, it was quite difficult to understand, review and reuse. So I switched to using Power BI instead, with notes against all the cards for ease of understanding.

Screenshot from the dashboard built for this (I have also attached the actual file):



Conclusion:

The probability analysis suggests that weather and congestion are **related, but only weakly**.

- **Not independent:** If weather and congestion were independent, clear weather and congestion would occur together in 4.96% of hours. The actual figure is 4.34%, so the two events are link ed.
- **Clear weather is slightly protective:** Clear hours account for 28.85% of congested hours, bel ow clear weather's overall share (32.94%). The odds of congestion in clear weather are about 26% lower than in cloudy weather (odds ratio 0.736), and 13.2% of clear hours are congested against 17.1% of cloudy hours.
- **Poorer visibility goes with more congestion:** Fog, clouds and mist have the highest congesti on rates (about 23%, 17% and 17%). Fog is based on only 201 hours, so read it with care.
- **High temperature tilts the other way:** High-temperature hours make up 27.82% of congested hours against 25.94% of all hours, a small di fference.
- **The effect is modest:** The differences are a few percentage points, so most congestion occurs whatever the weather. Weather is not a strong predictor of congestion on its own.
- **Association, not cause:** Other factors, such as time of day, may influence both weather and c ongestion.